

HMAC construction

$$H((K \oplus \text{opad}) || H((K \oplus \text{ipad}) || m))$$

HMAC uses a hash function
e.g. SHA256 with a secret
key and a 2 pass structure

$$HMAC(K, m) = H((K \oplus \text{opad}) || H((K \oplus \text{ipad}) || m))$$

- The key K is XORed with 2 padding constants - inner pad and outer pad
- An inner hash^{is} is computed over padded key + message
- An outer hash is then computed over the outer padded key + inner hash result
- The output is the authentication code, verified by any party holding K .

• Improvements over naive HMAC MAC

• $MAC(H(K || m))$;

• A simple hash of key || message is vulnerable to a length extension attack

• More Dangerous Hash functions (SHA-2, MD5) allow an attacker to know $H(k || m)$ to compute a valid Hash for $H(k || m || \text{extra})$ without knowing k .

HMAC defeats this because inner hash result is rehashed with a separate padded key, so attacker cannot extend it without knowing k .